

Yash Rawal

+91-9399159685 | yashrawal987@gmail.com | linkedin.com/in/rawal-yash | github.com/YashRL | Ratlam (457001), Madhya Pradesh, India

Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms. Specialized in PyTorch foundation model pretraining (~85M parameters), custom Triton/CUDA attention kernels, and self-evolving agent loops. Active contributor to Spring AI, bridging core machine learning research with high-performance production systems.

Skills

Languages & Backend: Python, Java, SQL, FastAPI, Spring Boot, PostgreSQL, REST APIs

AI & ML Engineering: Model Training & SFT, Agentic AI, Distributed Training & Inference, KV Caching, MCP, Vector Embeddings

AI & ML Frameworks: PyTorch, Triton, CUDA, HIP, vLLM, Hugging Face, Spring AI, LangGraph, CrewAI

Platforms & Models: Claude, Amazon Bedrock, Google Vertex AI, Azure OpenAI, Qwen, TinyStories, Meta Llama & Open Weights

Infrastructure & Security: Docker, Kubernetes, Linux, CI/CD, LLM-as-Judge, Red Teaming, PII Sanitization, WebSockets

Work Experience

Bigscal Technologies Pvt. Ltd.

AI/ML Engineer

Oct 2024 - Present

Remote

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by 70%.
- Trained domain-specific **Qwen2.5-7B** models on A100 GPUs and fine-tuned E5-large-v2 embeddings using PyTorch to optimize query intent mapping.
- Developed the Prompt Lab evaluation control plane to collect reasoning trajectories and human preference annotations, establishing a scalable **Human Feedback (RLHF) Pipeline**.
- Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
- Built production guardrails against prompt injections and data leaks, integrating runtime tool validation and **automated PII redaction**.

Bigscal Technologies Pvt. Ltd.

AI/ML Engineer Intern

Apr 2024 - Sep 2024

Surat, India

- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR (Whisper, Vosk, NeMo).
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

Projects

Mini Language Model (~85M Parameter Architecture) | Personal Research

Pretrained an ~85M-parameter transformer to ~1.43 validation loss on TinyStories; engineered FragmentStream chunked attention to eliminate $O(T^2)$ VRAM overhead on commodity GPUs and custom Triton kernels with KV caching for low-latency inference.

Python, PyTorch, CUDA, Triton, Tesla P100/T4

Modular Agent Harness & Memory System ("Brain") | Open Source

Principal Creator & Architect of an open-source agentic framework designed to automatically generate and evaluate high-performance CUDA/HIP workloads, implementing a formal ReAct loop (Thought-Action-Observation) with tactical tool routing.

Java, Spring Boot, PostgreSQL, WebSockets

Spring AI (spring-projects/spring-ai) | Open Source

Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.

Java, Spring Boot

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University

2022 - 2024

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)

Microsoft

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Microsoft